

Jerk Comparison for Rise Segment (0° to β)

Large spikes represent theoretically infinite jerk — the key failure mode in poor cam design

Normalised Jerk ($J / J_{\text{cycloidal_max}}$)

10

5

0

-5

-10

0

15

30

45

60

75

90

Cam Angle θ (degrees from start of rise)

Motion Type

— Uniform

· · · Parabolic

— Harmonic

- - - Cycloidal

8 jerk
(spike)

8 jerk
(parabolic
midpoint)

